

INVERSE PROBLEMS · HOLOGRAPHY · GENERATIVE AI

Solving Inverse Problems in Physics via AI

From Newton's force to the geometry of entanglement

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THE PROBLEM

Physics runs on inverse problems

Finding the law is only half the job. Inside a known law you still have to find the function that reproduces the data — the force, the potential, the metric. That target has grown:

$$\{c_i\} \longrightarrow V(x) \longrightarrow g_{\mu\nu}$$

a few constants $\rightarrow$ a whole function $V(x)$ $\rightarrow$ the geometry $g_{\mu\nu}$

Not unique

many laws fit the same data

Unstable

tiny noise $\rightarrow$ big change

No closed form

and hard at strong coupling

The inverse problem is ill-posed. Can AI solve it?

THIS TALK

Reconstruct the bulk geometry from entanglement

In $\text{AdS}_3/\text{CFT}_2$, the entanglement entropy is what we can measure; the bulk geometry stores all of that physics. Holography turns a brutal quantum computation into geometry.

- **Hard to compute**

EE is non-local — replica trick, and intractable at strong coupling.

- **RT: a sum becomes an area**

Ryu-Takayanagi turns EE into a minimal-surface area.

- **Depth = energy scale**

The radial direction is the RG scale, so EE encodes the UV→IR flow.

- **The payoff**

From the geometry: thermodynamics, the RG flow, strong-coupling physics.

SETUP

One unknown function, $f(z)$

THE METRIC

$$ds^2 = \frac{L^2}{z^2} \left[-f(z) dt^2 + \frac{dz^2}{f(z)} + dx^2 \right]$$

$$f(z)$$

the blackening function — all the physics lives here

$$f(0) = 1$$

boundary condition — asymptotically AdS

$$f(z_h) = 0$$

a horizon — a black-hole (thermal) geometry

THE FORWARD MAP

the inverse, $S(\ell) \rightarrow f(z)$, has no closed form — this talk

$$S(\ell) = \frac{\text{Area}(\gamma_A)}{4 G_N} \longrightarrow S(\ell) = \frac{L}{4 G_N} \int_{-\ell/2}^{\ell/2} \frac{dx}{z} \sqrt{1 + \frac{(z')^2}{f(z)}}$$

APPROACH 1

Physics-informed neural network

Park, Hwang, Cho & Kim — arXiv:2205.04445

Fit the geometry as a network

Let the unknown geometry be a neural network, and adjust its weights until the entropy matches the data.

$$f(z) = f_{\theta}(z)$$

$$\text{Loss}(\theta) = \sum_i |S_{\text{RT}}(z_t^i) - S_{\text{data}}(\ell(z_t^i))|$$

S_{RT}

gravity side

S_{data}

quantum (CFT) data

RECOVERS · WITH ITS $f(z)$

- **Black hole**

$$f(z) = 1 - \left(\frac{z}{z_h}\right)^2$$

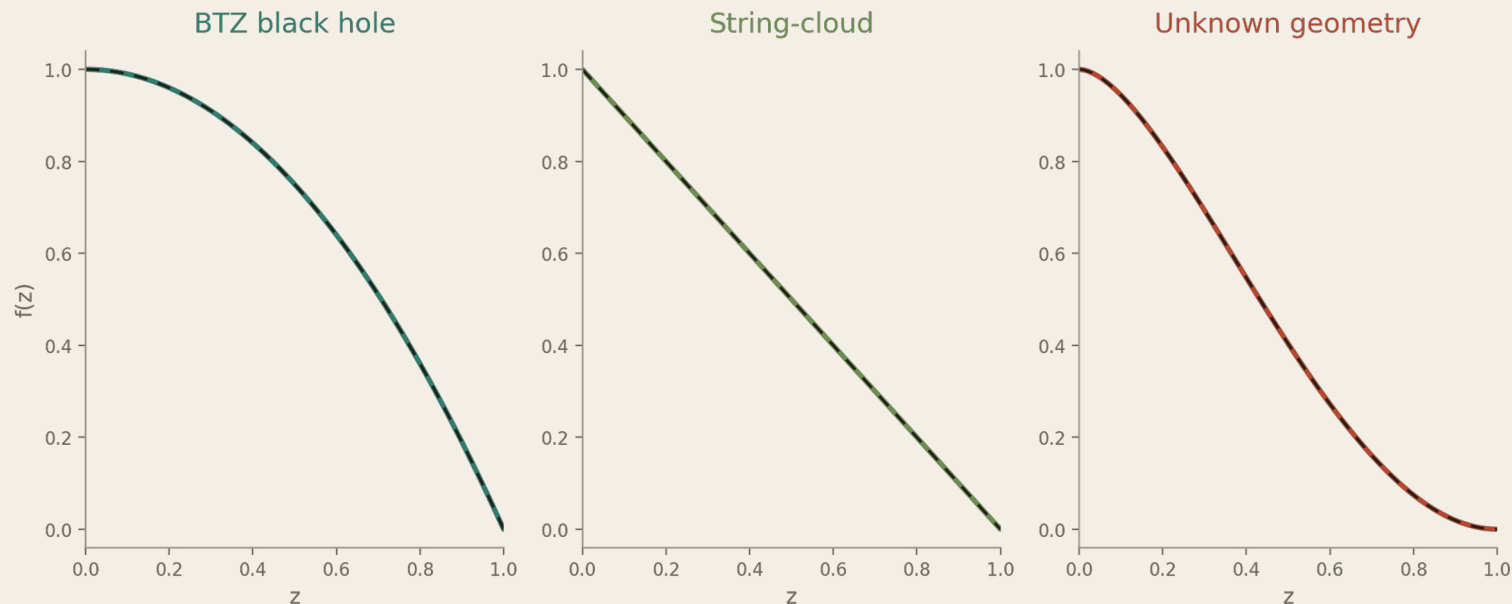
- **String cloud**

$$f(z) = 1 - z/z_h$$

- **Unknown geometry**

$$f(z) = ?$$

Recovered — but one training, one solution



dashed = reconstruction · solid = truth

THE LIMIT

One network is fit to one entropy curve. A new theory means a new loss and a fresh training. No transfer, no parameter sweep.

APPROACH 2 · THIS WORK

A generative model: the Transformer

Sejin Kim — arXiv:2511.06387

From entropy data to model input

HAMILTON-JACOBI RELATION

$$\frac{dS}{d\ell} = -H(z_t)$$

the left side comes from the data, the right side from the geometry

- The minimal surface turns around at its deepest bulk point z_t , where $dz/dx = 0$.
- Translation symmetry along x gives a conserved quantity $H = -L/(4G_N z_t)$, fixed by z_t alone.
- $S(\ell)$ is measured, so the left side is known, and inverting the relation yields the input $\ell(z_t)$.
- This step is computed in advance, and $f(0) = 1$ keeps the inverse well-defined.

Learn the inverse map, once

$$\text{Transformer}(x_{1:n}, y_{<t}) = y_t$$

Treat data and geometry as sequences and learn the map with a Transformer. We train the model's weights, not the physics — so inference needs no per-problem loss, and one training solves many problems.

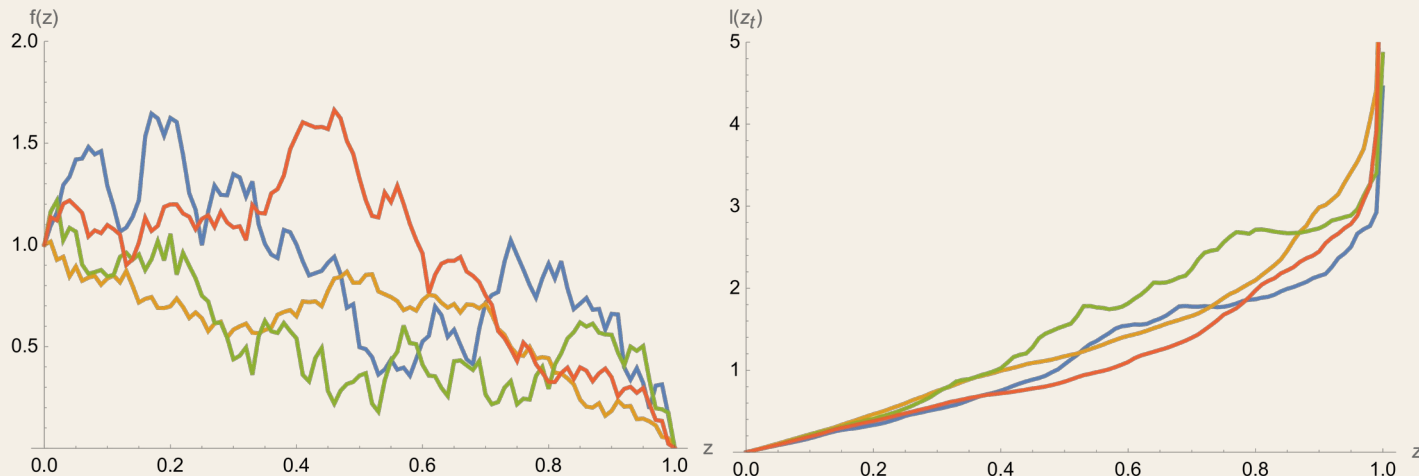
source → **target**

$$\{\ell_0, \dots, \ell_N\} \mapsto \{f_0, \dots, f_N\}$$

$\ell(z_t)$ as input, $f(z)$ as output

Noise makes the model feel the physics

$$f(z) = 1 - (z/z_h)^2 + \eta(z_h - z)$$



random noisy $f(z) \rightarrow$ boundary data $l(z_t)$ from RT

NUMBERS

$\sigma = 0.5$

noise strength

$\Delta z = 0.01$

grid

101

seq. length

100k

pairs · 80/20

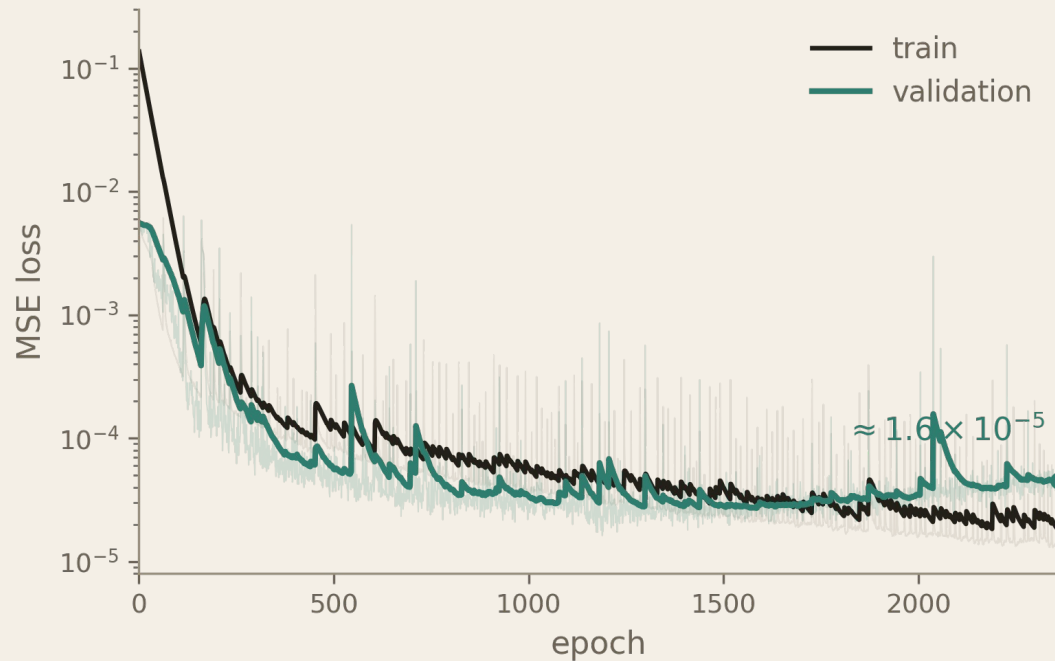
$l \leq 10, f \leq 2$

value range

The noise vanishes at the horizon, so $f(z_h) = 0$ stays exact.

OPTIMIZATION

Trained once — then frozen



RECIPE

Adam · MSE

optimizer · loss

2.4k epochs

batch 256

1.6×10^{-5}

best validation loss

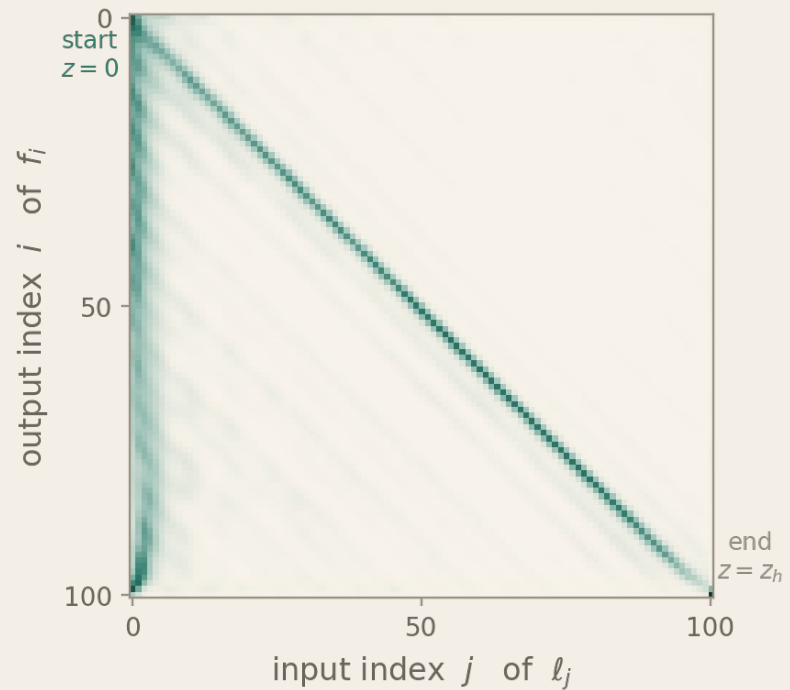
$\approx 12\text{M}$ params

3+3 enc/dec layers

The weights are frozen after this single run — every result in the rest of this talk comes from this one checkpoint.

WHAT IT LEARNS

Reads up to its own depth



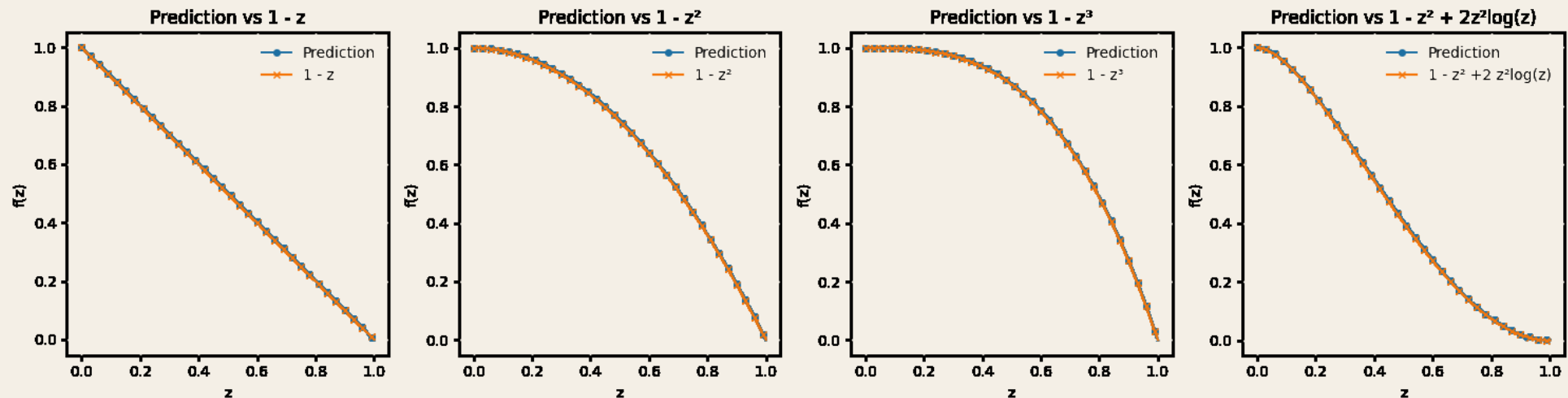
cross-attention · best.pth

- The upper triangle is almost empty, so output f_i barely uses input values at larger index j .
- So f at z_i is fixed by the entanglement data up to z_i and the part of f already produced — the causal structure of the inverse problem.
- The small residual is a finite-training effect; with more training the upper triangle should converge to zero.

RESULT 1 OF 5 · SMOOTH TEST

Smooth black holes it never saw

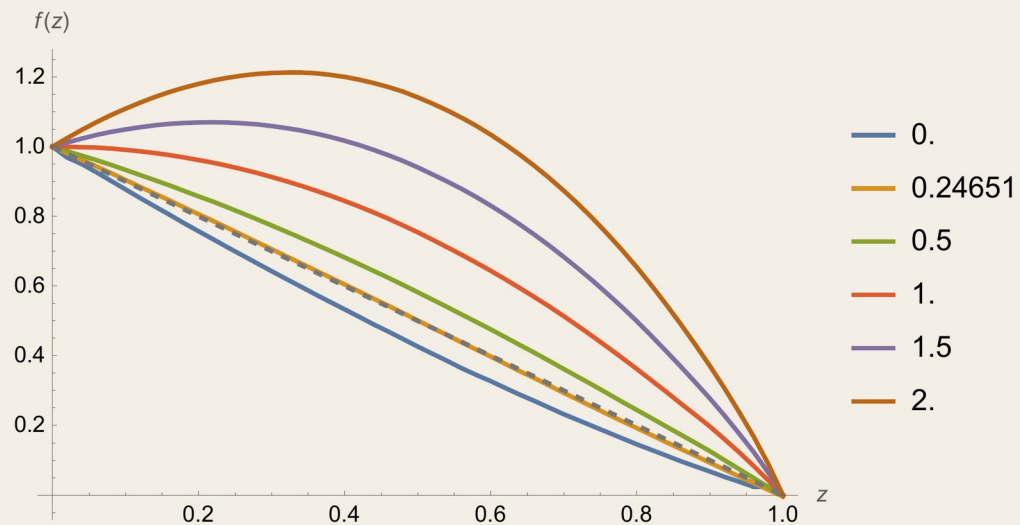
Trained only on noisy $1 - z^2$. Tested on four clean profiles — dots = prediction, line = truth.



Why it matters: clean curves emerge from noisy data, and charged BTZ ($1 - z^2 + 2z^2 \log z$) is a different family. The model learned the RT integral, not the BTZ family.

A whole family, swept in one model

$$S(\ell; s) = \frac{L}{2G_N} \log \left[\frac{2z_h}{\epsilon(s+1)} (e^{\ell/2z_h} - e^{-s\ell/2z_h}) \right]$$

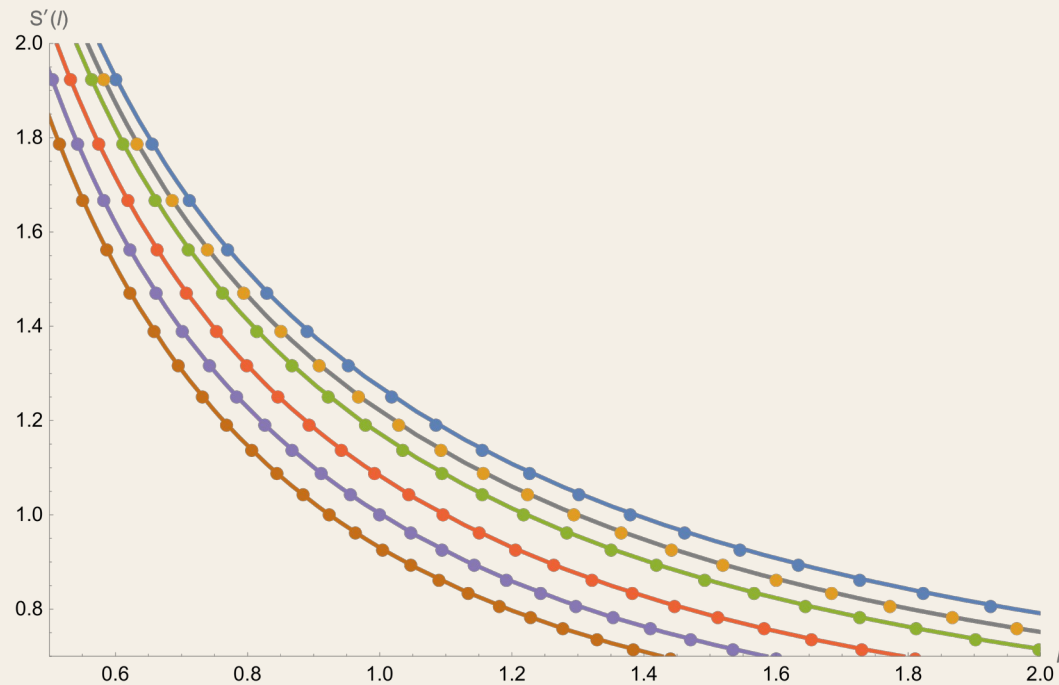


WHAT HAPPENS

- One model predicts $f(z;s)$ across the whole parameter s .
- At $s = 0.24651$ the entropy happens to match the p-brane gas, and the prediction comes out near $f = 1 - z$.

CONSISTENCY CHECK

Predictions close the loop



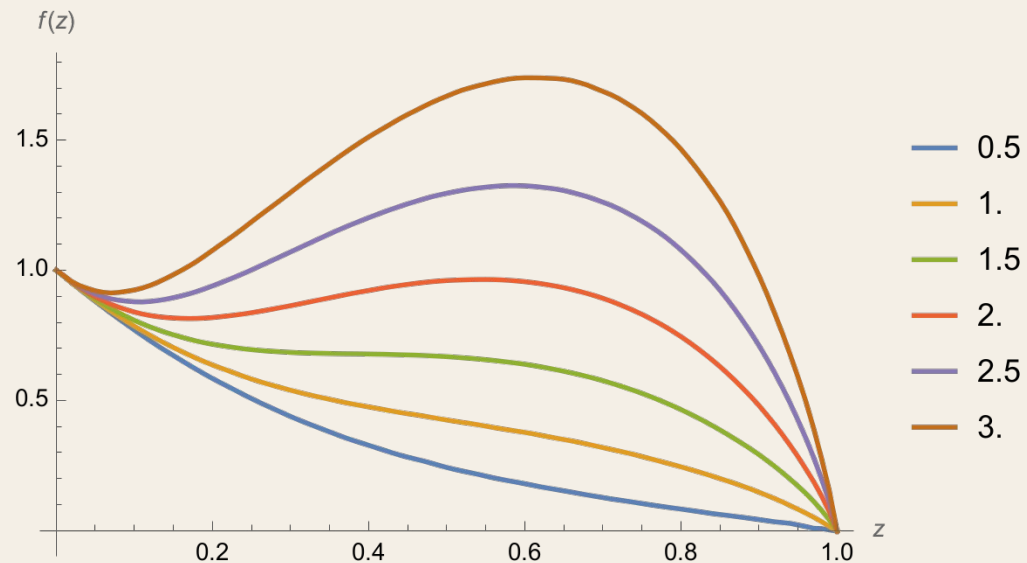
$S'(l)$: input (solid) vs recomputed (dotted)

ROUND TRIP

- Insert the predicted $f(z)$ into the exact RT integral and recompute $S'(\ell)$.
- The recomputed $S'(\ell)$ lies on the input data for the whole family.
- The test needs no true answer, so it can run on real data.

The tanh family, same weights

$$S(\ell; s) = \frac{L}{2G_N} \log \left[\frac{2z_h}{\epsilon_{UV} s} \tanh\left(\frac{s\ell}{2z_h}\right) e^{\ell/(2z_h)} \right]$$



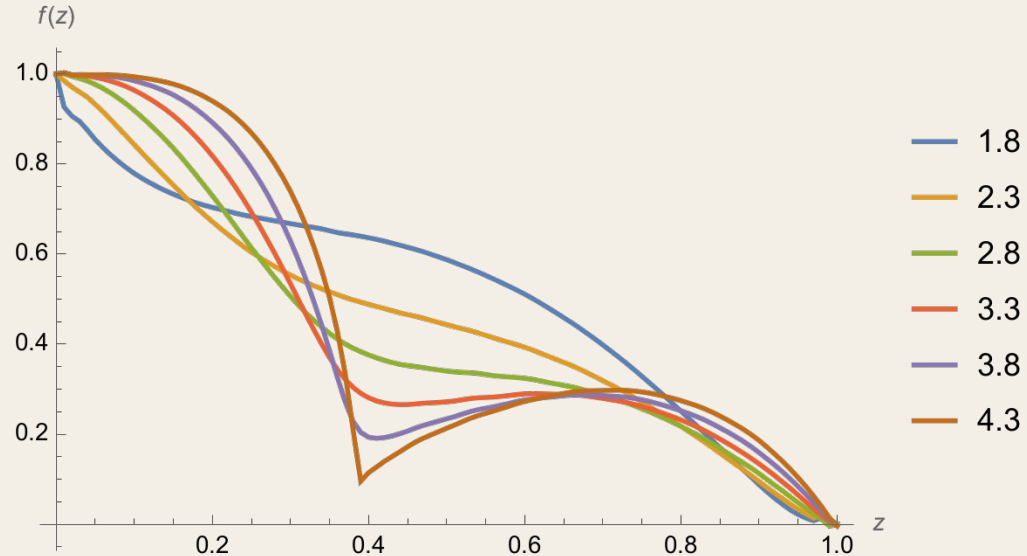
predicted $f(z;s)$, $s \in \{0.5, \dots, 3\}$

WHAT HAPPENS

- Non-monotonic profiles with a growing bump — far from the training family.
- Valid for $0.5 \leq s \leq 3$ — where f and ℓ stay inside the trained range.

Power interpolation, at the edge

$$S(\ell; s) = \frac{L}{2G_N} \log \left[\frac{2z_h}{\epsilon_{UV}} \frac{\frac{\ell}{2z_h} + e^{\ell/(2z_h)} \left(\frac{\ell}{2z_h}\right)^s}{1 + \left(\frac{\ell}{2z_h}\right)^s} \right]$$



predicted $f(z;s)$, $s \in \{1.8, \dots, 4.3\}$

WHAT HAPPENS

- Sharp, near-singular features develop as s grows.
- Such sharp features are hard to reconstruct — an honest limit of the method.
- Beyond $s \approx 4.3$ the predicted f touches zero inside the bulk — a new horizon forms.

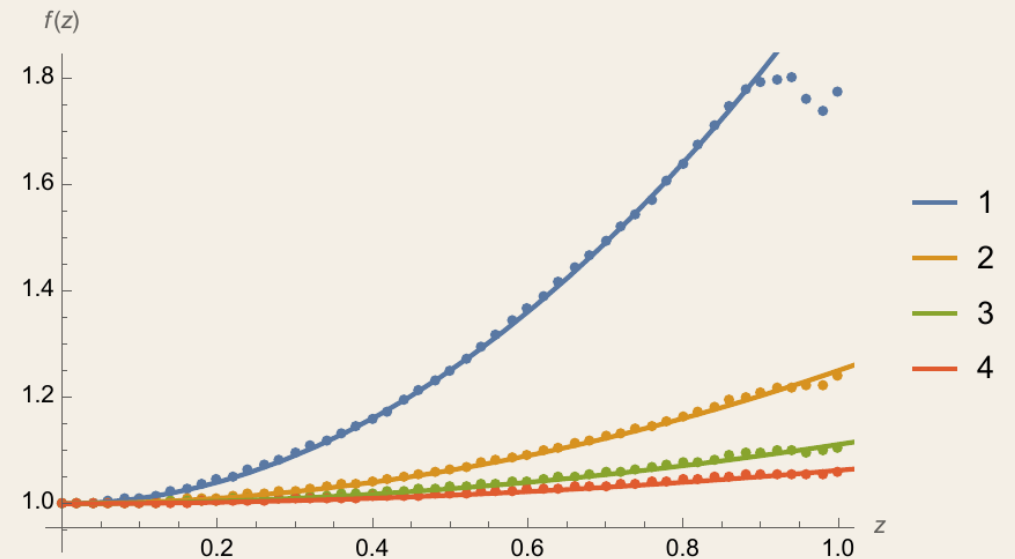
Beyond the horizon — literally

$$S(\ell) = \frac{L}{2G_N} \log\left(\frac{2s}{\epsilon} \sin \frac{\ell}{2s}\right)$$

$$f(z) = 1 + (z/s)^2$$

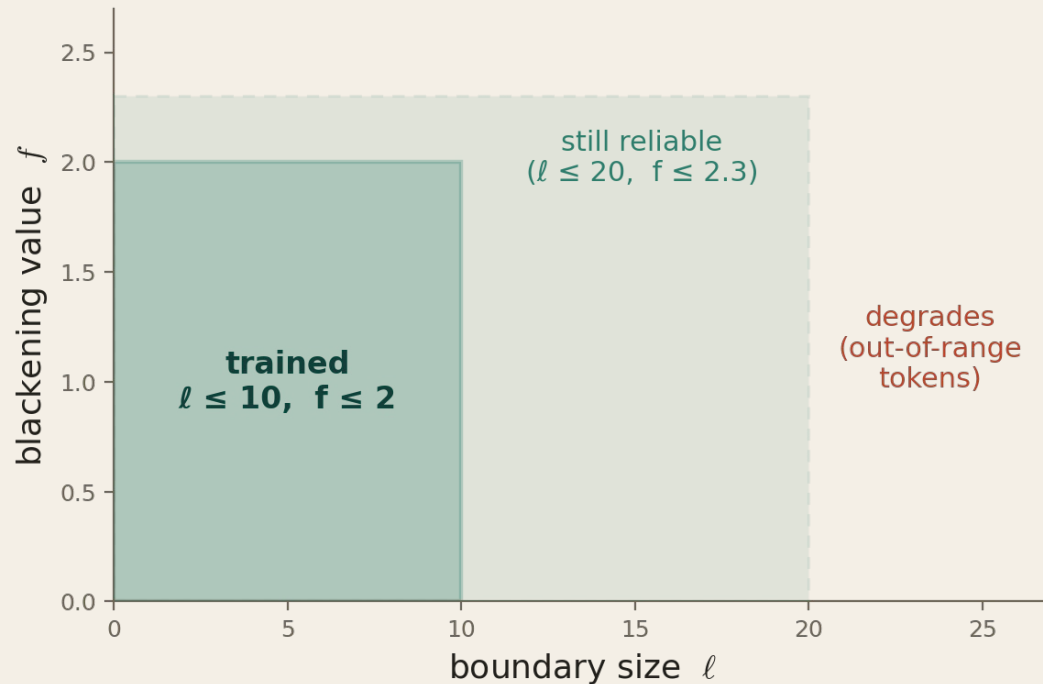
Periodic-boundary entropy → a geometry with no horizon at all.

Trained only on horizon geometries $f(z_h) = 0$ — yet it reconstructs horizonless ones too, with only mild instability near $z = 1$.



HOW FAR IT WORKS

The window of validity



trained range and the range where predictions stay reliable

- Training covered ℓ up to 10 and f up to 2, and predictions stay reliable to about $\ell = 20$ and $f = 2.3$.
- The values act as tokens, so values far outside the trained range have no learned representation.
- The tanh family stops at $s = 3$, where f leaves the range, and the horizonless test pushes past it near $z = 1$.

Outside the window the model fails visibly, not silently.

WHY IT COULD EXTRAPOLATE

It learned the relation, not the examples

Trained on one geometry, it answered correctly for many it had never seen. That happens only if it understood the integral relation.

- **Learned the map, not a curve** It captured how ℓ depends on $f(z)$ point by point — the RT integral.

- **$f(z)$ can be anything** In the math, $f(z)$ is a free function with no fixed form.

- **So new cases just work** Smooth, charged, even horizonless geometries follow from the same relation.

Solver vs. learned map

Physics-informed NN

- Fits one entropy curve
- New loss per theory
- ✓ Full geometry + thermodynamics
- Retrain for every system

Generative Transformer

- ✓ Learns the inverse map
- ✓ One training, many problems
- ✓ No per-problem loss
- ✓ Generalizes to unseen geometries

What we actually did

- **Reconstructed spacetime** from a purely quantum quantity — entanglement
- **AI as a physics solver** learning the inverse of a hard integral operator
- **A reusable map** one model transfers across theories, not one fit per problem

HONESTLY

Limits & what's next

LIMITS

- Reliable only inside the trained token range
- Noise can break energy conditions — a math solver, not a physical one
- AdS_3 only; sharp or singular geometries stay hard

NEXT

- Time-dependent entropies and other region shapes
- Broader QFT families and higher dimensions
- Generative inversion as a general tool for physics

Thank you for your attention

From finding forces to reconstructing spacetime — AI is learning to invert the laws of physics.

Park, Hwang, Cho & Kim · [arXiv:2205.04445](#)

Sejin Kim · [arXiv:2511.06387](#)

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